



16 August 2024

OTHER THAN FULL AND OPEN COMPETITION JUSTIFICATION AND APPROVAL

OTHER THAN FULL AND OPEN COMPETITION JUSTIFICATION

1. Nature and/or Description of Action

The Defense Threat Reduction Agency contracting activity proposes to procure without using full and open competition a void sensing fuze algorithm developed and used by Northrop Grumman on its penetrator weapons systems. The scope of this contract includes integrating proprietary algorithms into the Integrated Munitions Effect Assessment (IMEA) software tool to meet the IMEA and Target Weaponeeing Assistance Cell (TWAC) programs' technical objectives. The fuzing "black box" algorithm will be used on IMEA and TWAC weapons analysis efforts. The government will gain limited rights to this proprietary product for the duration of the Period of Performance (PoP). The algorithm will not be altered or further developed by the government team. This contract will not be for Advisory and Assistance Services.

2. Description of the Supplies/Services Required

The purpose of this contract is to acquire a set of proprietary code for improving the accuracy of HTD weapons modeling in the Integrated Munitions Effect Assessment (IMEA) software tool. The code has already been developed by Northrop Grumman, its architecture based off the IMEA tool structure and for the program's intended use cases. This code has already been used to generate modeling results for the TWAC and IMEA programs, who have verified its effectiveness in meeting technical objectives. Finding another vendor to produce this code would create duplication of effort and waste a significant amount of time as the vendor would start development back at square one.

Furthermore, the Hard Target Void Sensing Fuze, the smart fuze technology that is used in the weapons that the TWAC and IMEA teams need to replicate in modeling and simulation software, is owned proprietary by Northrop Grumman. This means in order for another company to develop the code required to simulate these HTD weapons, Northrop Grumman would have to share its proprietary smart fuze technology with them, which is an impossibility. Thus, a contract is needed for this mission to be accomplished and for TWAC to continue providing weaponeeing recommendations to Air Force operators and Combatant Commands (COCOMs). With these considerations, there are no other contractors capable of meeting the Government's need. Therefore, based on the information provide above, only one responsible source will satisfy the agencies requirements.

The desired outcome to this contract is to accurately predict the timing of the void sensing fuze technology used in penetrator weapons Modeling & Simulation (M&S) results from the IMEA software tool. The work required to meet the Agency's needs is integration, documentation, and

validation of the smart fuze algorithm into the IMEA software tool. This is new work to be performed on an existing software. The total PoP is 8 months. The base period is 4 months followed by 3 option periods that are 4 months, 3 months and 3 months, respectively. Some option periods will start work and run concurrently with the execution of other phases, resulting in total Period of Performance on the effort being 8 months. [REDACTED]

3. Statutory Authority

The statutory authority for this non-competitive action is 10 United States Code (U.S.C.) 3204(a)(1), as implemented by FAR 6.302-1, Only One Responsible Source, and no other services will satisfy agency requirements.

4. Applicability of Authority

Northrop Grumman is the only responsible source capable of providing this deliverable without unacceptable interruption and degradation of SE&I and SME support to operational units resulting in a significant delay in meeting requirements by CCMD and Intelligence Community users of the IMEA software. [REDACTED]

Per FAR 6.302-1(a)(2) when the supplies or services required by the agency are available from only one responsible source, or, for DoD, NASA, and the Coast Guard, from only one or a limited number of responsible sources, and no other type of supplies or services will satisfy agency requirements, full and open competition need not be provided for (i) Supplies or services may be considered to be available from only one source if the source has submitted an unsolicited research proposal that (A) Demonstrates a unique and innovative concept (see definition at 2.101), or, demonstrates a unique capability of the source to provide the particular research services proposed.

This applies to this program as the services being rendered and product being delivered includes proprietary algorithms developed and owned by Northrop Grumman that are uniquely required for simulating the smart fuze systems on the weapons to be modeled in the IMEA software. No other source can provide this proprietary algorithm without extensive duplication costs being incurred by the government and creating significant schedule delays in meeting agency requirements.

5. Effort to Solicit Potential Sources

Since Northrop Grumman is the original developer of the smart void sensing fuze algorithm in place in current weapons systems, the very weapons systems the IMEA and TWAC programs hope to accurately model & simulate in the IMEA software, they are the only source that can meet this requirement as the algorithm is custom developed and proprietary.

As such, a Sources Sought Notice was not published, however, a Notice of Contract Action will be published as required by FAR 5.201.

6. Fair and Reasonable Costs

The anticipated cost/price of this contract action to the Government will be determined to be fair and reasonable by the Contracting Officer based upon analysis of the proposal to an Independent Government Cost Estimate (IGCE) built on historical cost information of similar supplies/services, inputs and evaluations from the DTRA Program Office, and analysis of cost and pricing data, in accordance with cost analysis techniques found in FAR 15.404-1(c).

7. Market Research

The desired non-commercial deliverable is only available via a specific, sole source contractor for this acquisition. The smart fuze technology (Hard Target Void Sensing Fuze,) that is used in the weapons that the TWAC and IMEA teams need to replicate in modeling and simulation software, is owned proprietary by Northrop Grumman. This means in order for another company to develop the code required to simulate these HTD weapons, Northrop Grumman would have to share its proprietary smart fuze technology with them, which is an impossibility. Thus, a contract is needed for this mission to be accomplished and for TWAC to continue providing weaponeering recommendations to Air Force operators and Combatant Commands (COCOMs). With these considerations, there are no other contractors capable of meeting the Government's need. Therefore, based on the information provide above, only one responsible source will satisfy the agencies requirements.

8. Other Facts

Estimated cost duplication to the government would be \$1.5M. This estimate was generated using historical acquisition information of similar supplies/services. Development of smart fuze algorithms by another company would be very costly in both time and money.

9. Interested Sources

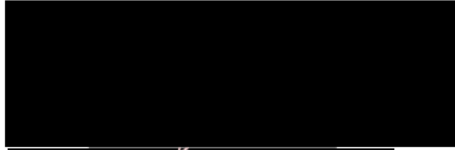
None.

10. Subsequent Actions

At present, Northrop Grumman is the only known source with the depth of knowledge and experience, proprietary algorithms, and demonstrated performance-based capability to fulfill these unique requirements. Pursuing contract work with another vendor for this effort will cause significant duplication costs to be incurred by the government as well as unacceptable delays in the timeline for work to be done.

Technical Certification

I certify that the data and information forming the basis for this justification are accurate and complete to the best of my knowledge and belief.



16 Aug 2024
Date

Program Manager/Project Officer

Contracting Officer Certification and Determination

I certify that this justification is accurate and complete to the best of my knowledge and belief.

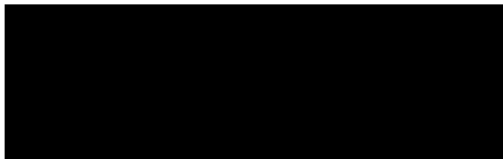


16 Aug 2024
Date

Contracting Officer

General Counsel Coordination

This justification is legally sufficient.



16 Aug 2024
Date

Attorney